

The body's muscles

FROM A LION GNAWING AT A GAZELLE to a human eating a steak, a chunk of juicy muscle is the predator's prime meal in the natural world. On average most animals, including humans, are one-half muscle. Muscles are the body's movers. This is because a muscle is a lump of tissue specialized to do a simple job: to shorten, or contract.

Every movement, from the blink of an eye to sprinting in a race, is driven by the body's muscular system.

Muscles, along with the bones they pull, were among the first parts of the body to be drawn, described, and named, because of the general interest in the body's movement – and in eating meat. Galen of ancient Rome identified about 300 individual skeletal muscles, just under half the body's total. Leonardo da Vinci noted and named more, and his extraordinary engravings and paintings of living people reflected this knowledge in the supple bulges and curves under their skin. The volume *Epitome* of Andreas Vesalius's monumental work *De Fabrica* (1543) is a classic of science and art, depicting yet more muscles in carefully dissected and posed bodies.



THE ULTIMATE BOOK

Italian anatomist and teacher Giorgio Baglivi (1668-1707) told his students: "You will never find a more interesting, more instructive book than the patient himself." He had an overly simple belief that the body was a set of small machines. He regarded the blood vessels as a system of water pipes and the lungs as bellows. However, he was the first to note that skeletal muscles, which pull on bones, are different from muscles that work the intestines and other organs. Under the microscope, skeletal muscles look striped, while gut muscles do not.

THREE TYPES OF MUSCLE

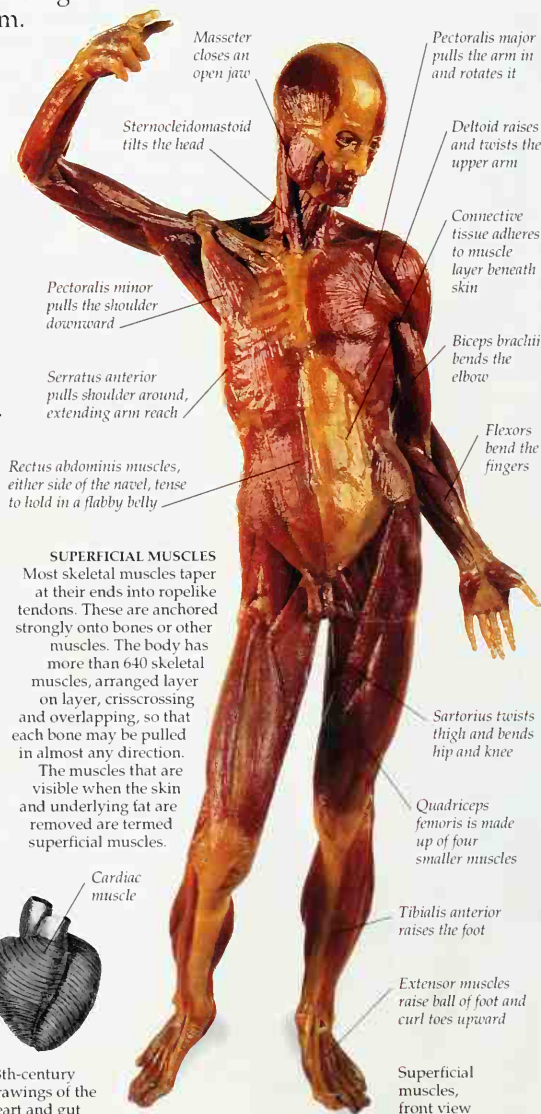
The body's three types of muscle have a variety of names. One type is called skeletal muscle, because it moves the skeleton's bones; it is also called striped or striated muscle (see drawing, top right) – under the microscope it has a striped appearance – and voluntary muscle, as it responds to the brain's conscious will. Visceral muscle in the intestines and abdominal organs is called smooth muscle, because it lacks stripes, and involuntary muscle, as it works without conscious involvement. The third type is cardiac muscle or myocardium, found solely in the heart.



18th-century drawings of the heart and gut



ROCKS AND MUSCLES
Danish scientist Niels Stensen (1638-1686) was a student at Copenhagen, Amsterdam, and Leiden. He carried out microscopic work on muscles and realized their contraction was the result of the combined shortening of thousands of tiny, thin fibers that make up each whole muscle. Besides his five books on anatomy, he was a crystal expert, and the "father" of geology.



SUPERFICIAL MUSCLES

Most skeletal muscles taper at their ends into ropelike tendons. These are anchored strongly onto bones or other muscles. The body has more than 640 skeletal muscles, arranged layer on layer, crisscrossing and overlapping, so that each bone may be pulled in almost any direction.

The muscles that are visible when the skin and underlying fat are removed are termed superficial muscles.